University College London, 2025-2026
Econ 0107 – Macroeconomics I – Łukasz Rachel & Franck Portier

Midterm Assessment

Answer the four problems. I: 30% of the total mark, II: 20% of the total mark, III and IV: 25% of the total mark each.

I – Ricardian Equivalence

Consider a two-period economy populated with a continuum of measure 1 of identical households. There is unique non storable good. Preferences of household i are

$$U = \log c_{i1} + \beta \log c_{i2}$$

with $\beta \in]0,1[$.

Each household receives endowments $\omega_1 > 0$ in period 1 and $\omega_2 > 0$. They have access to bonds, denoted B_i with (gross) interest rate R.

There is a government who taxes endowments in the periods at rates $\tau_1 > 0$ and $\tau_2 > 0$ and distribute a check $\theta > 0$ in period 1. Taxes and checks are the same for all then agents. The government can also issue debt B.

1 − Write and solve for the optimal problem of a representative household

Solution: Budget constraints are

$$\begin{cases} B_i + c_{1i} & \leq & (1-\tau_1)\omega_1 + \theta, \\ c_{i2} & \leq & (1-\tau_2)\omega_2 + RB_i, \end{cases}$$

so that the intertemporal budget constraint is

$$c_{i1} + \frac{c_{i2}}{R} = \underbrace{(1-\tau_1)\omega_1 + (1-\tau_2)\frac{\omega_2}{R} + \theta}_{\Omega_i}.$$

Euler equation is

$$\frac{c_{i2}}{\beta c_{i1}} = R.$$

Utility maximization gives optimal choices:

$$c_{i1} = \frac{1}{1+\beta}\Omega_i,$$

$$c_{i1} = \frac{\beta}{1+\beta}\frac{\Omega_i}{R}.$$

2 − Write the government period 1, period 2 and intertemporal budget constraints.

Solution:

$$\begin{cases} \text{Period 1} & : & \theta & \leq & \tau_1\omega_1 + B, \\ \text{Period 2} & : & RB & \leq & \tau_2\omega_2, \end{cases}$$

so that the intertemporal budget constraint is

$$\theta = \tau_1\omega_1 + \frac{\tau_2\omega_2}{R}.$$

3 – Define a competitive equilibrium

Solution: A competitive equilibrium is a budget feasible government plan (τ_1, τ_2, θ), a consumption plan (c_{i1}, c_{i2}) for echat agent i and a (gross) interest rate R such that

1. each household optimally chooses (c_{i1}, c_{i2}) when taxes, checks and prices are (τ_1, τ_2, θ) and R.
2. All markets clear (bonds and goods in both periods).

4 – Solve for competitive equilibrium price and quantities

Solution: In equilibrium, good market clearing implies $c_1 = \omega_1$ and $c_2 = \omega_2$. Euler equation then imply $R = \frac{\omega_2}{\beta\omega_1}$.

5 – Assume that the government increases the check distributed in period 1 and finances it by increasing τ_2 while leaving τ_1 constant. How does this affect price, bonds holdings and consumption quantities? Explain.

Solution: Quantities and prices are not affected. This is Ricardian equivalence. As the government has to finance the check in period 1, it uses debt, and therefore households save the check they are receiving to pay for the increase in taxes in period 2.

⋆

Let us now assume that a mass $\delta \in]0, 1[$ of agents dies at the end of period 1, and that they are informed of their death at the beginning of period 1 before taking any decision. These agents are denoted the *unlucky agents* "u". They are replaced in period 2 by a mass δ of *newborn* ("b"). Finally, a mass $1 - \delta$ of agents live for the 2 periods (the *lucky ones* "ℓ"). All the agents alive in a given period receive the same endowments, the same check if in period 1 and pay the same taxes. Utilities are $\log c_{\ell 1} + \beta \log c_{\ell 2}$ for the lucky agents, $\log c_{u1}$ for the unlucky ones and $\beta \log c_{b2}$ for the newborn.

6 – Solve for the optimal behaviour of the three types of agents.

Solution: Given taxes, check and R, the budget constraints and optimal behaviour of the lucky agents is the same as the one of the households in the previous questions. Unlucky and newborn agents make no choice but consuming all their income as they live only for one period. Their budget constraints are

$$c_{u1} = (1 - \tau_1)\omega_1 + \theta,$$

and

$$c_{b2} = (1 - \tau_2)\omega_2.$$

7 – Define a competitive equilibrium

Solution: A competitive equilibrium is a budget feasible government plan (τ_1, τ_2, θ), a consumption plan (c_{i1}, c_{i2}) for each household ℓ, u, b and a (gross) interest rate R such that

1. ℓ households optimally chooses $(c_{\ell 1}, c_{\ell 2})$ and u, b households spend all their resources in consumption when taxes, checks and prices are (τ_1, τ_2, θ) and R.

2. All markets clear (bonds and goods in both periods).

8 – Assume that the government increases the check distributed in period 1 and finance it by increasing τ_2 while leaving τ_1 constant. How does this affect price and quantities? Explain.

Solution: Consumptions of the various households types are

$$c_{\ell 1} = \frac{1}{1+\beta}\Omega_\ell, \tag{$\star$}$$

$$c_{\ell 2} = \frac{\beta}{1+\beta}\frac{\Omega_\ell}{R},$$

$$c_{u1} = (1-\tau_1)\omega_1 + \theta,$$

and

$$c_{b2} = (1-\tau_2)\omega_2.$$

Government budget constraint is

$$\theta = \tau_1\omega_1 + \frac{\tau_2\omega_2}{R},$$

and good market equilibrium conditions in the two periods are

$$(1-\delta)c_{\ell 1} + \delta c_{u1} = \omega_1, \tag{$\star\star$}$$

$$(1-\delta)c_{\ell 2} + \delta c_{b2} = \omega_2.$$

Using ($\star$) and ($\star\star$), we obtain

$$R = \frac{\omega_2}{\beta\omega_2}\left(1 + \frac{\delta(1+\beta)\tau_2}{1-\delta}\right).$$

Note that when $\delta = 0$, we are back to the previous case, and $R = \frac{\omega_2}{\beta\omega_2}$. We see that an increase in θ, which implies an increase in τ_2, will imply increase in the real interest rate. Also note that $\frac{\partial c_{u1}}{\partial\theta} > 0$. Because unlucky agents will not suffer from a future increase in taxes, the check comes as net wealth, and they increase their consumption. As total consumption in period 1 is constant (ω_1, the lucky agents will decrease their consumption in period 1. c_{b2} will decrease as τ_2 increase. As total consumption in period 2 is constant (ω_2), $c_{\ell 2}$ will increase.

Allocations are non Ricardian because the unlucky agents receive the check but will never pay the future taxes that finance that check.

II – Two Agents with Complete Markets and Redundant Asset

An economy consists of two infinitely lived consumers named $i = 1, 2$. There is one nonstorable consumption good. Consumer i consumes c_t^i at time t. Consumer i ranks consumption streams by

$$\sum_{t=0}^{\infty} \beta^t u(c_t^i),$$

where $\beta \in (0, 1)$ and $u(c)$ is increasing, strictly concave, and twice continuously differentiable. Consumer 1 is endowed with a stream of the consumption good

$$y_t^1 = 1, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0 \dots$$

Consumer 2 is endowed with with a stream of the consumption

$$y_t^2 = 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, \dots$$

Assume that there are complete markets with time-0 trading.

1 – Define a competitive equilibrium.

Solution: A competitive equilibrium is a feasible allocation $\{c_t^i\}_{t=0}^{\infty}$ for each agent $i = 1, 2$ and a price sequence $\{q_t^0\}_{t=0}^{\infty}$ such that:

1. Given prices, the allocation solves the household's problem for each i.
2. The allocation is feasible: $\sum_i c_t^i = \sum_i y_t^i$.

2 – Compute a competitive equilibrium.

Solution: The first-order conditions for optimality of the household problem imply for each agent $i = 1, 2$:

$$q_t^0 = \beta^t \frac{u'(c_t^i)}{u'(c_0^i)}.$$

Let us guess and verify a competitive equilibrium in which the consumption of each agent is constant across time. The first order condition implies then that $q_t^0 = \beta^t$. To find c_i, use agent i's budget constraint :

$$(1 - \beta) \sum_{t=0}^{\infty} \beta^t y_t^i = c^i.$$

For agent 1 this gives:

$$c_1 = (1 - \beta)(1 + \beta + \beta^4)(1 + \beta^6 + \beta^{12} + \beta^{18} \cdots) = \frac{(1 - \beta)(1 + \beta + \beta^4)}{1 - \beta^6} = \frac{1 + \beta + \beta^4}{1 + \beta + \beta^2 + \beta^3 + \beta^4 + \beta^5}.$$

For agent 2 market clearing implies that $c_2 = 1 - c_1$.

3 – Suppose that one of the consumers markets a derivative asset that promises to pay $0 < d < 1$ units of consumption in periods $0, 2, 4, \dots$ and $-d$ in periods $1, 3, 5, \dots$. What would the price of that asset be?

Solution: Since markets are complete, the derivative security is redundant: it can be priced off the Arrow-Debreu priced determined in question 2. The time zero price of the derivative is

$$\begin{aligned} P_0 & \quad \sum_{t=0}^{\infty} q_t^0 d \\ &= \sum_{t=0}^{\infty} \beta^{2t} d - \sum_{t=0}^{\infty} \beta^{2t+1} d \\ &= \sum_{t=0}^{\infty} \beta^{2t} d - \beta \sum_{t=0}^{\infty} \beta^{2t} d \\ &= (1-\beta) d \left(\sum_{t=0}^{\infty} \beta^{2t} \right) \\ &= \frac{(1-\beta)d}{1-\beta^2} \\ &= \frac{d}{1+\beta}. \end{aligned}$$

III – Stochastic growth with elastic labor supply

Suppose the planner seeks to maximize

$$E\left\{ \sum_{t=0}^{\infty} \beta^t u(c_t, l_t) \right\}, \qquad 0 < \beta < 1,$$

subject to the resource constraint

$$c_t + k_{t+1} = z_t f(k_t, l_t) + (1-\delta) k_t, \qquad 0 < \delta < 1,$$

with initial conditions $k_0 > 0$ and $z_0 > 0$. Productivity z_t evolves according to a Markov process $F(z'|z)$.

The planner chooses (c_t, l_t). Utility is increasing and concave in c, decreasing and convex in l. Production is increasing, concave, and CRS.

1 – Let $v(k, z)$ denote the planner's value function. Set up the Bellman equation.

Solution: The Bellman equation is

$$v(k, z) = \max_{c, l, k'} \left[u(c, l) + \beta \int v(k', z') \, dF(z'|z) \right]$$

subject to the resource constraint

$$c + k' = z f(k, l) + (1-\delta) k.$$

The Bellman equation characterizes the value of starting a period with capital k and productivity z, choosing labor l, consumption c, and next period's capital k', and then continuing optimally.

2 – Discuss briefly how the contraction mapping theory helps us solve this problem. In particular, is the solution v unique? What condition, if any, would guarantee that the optimal plan is unique?

Solution: Contraction Mapping Theorem says that if an operator is a contraction mapping, then it has a unique fixed point and converges from anywhere. So we can use VFI to find the solution. (Blackwell Sufficiency Theorem tells us whether a mapping is a contraction mapping (needs to satisfy monotonicity and discounting). If v is strictly concave, then the plan is unique. This will be the case is u is strictly concave.

3 – Derive the planner's optimality conditions for c, k', and l.

Solution: Substitute $c = zf(k,l) + (1-\delta)k - k'$ into the objective. The RHS becomes

$$u(zf(k,l) + (1-\delta)k - k',\, l) + \beta \int v(k', z')\, dF(z'|z).$$

FOC for labor:

$$u_c(c,l)\, z f_l(k,l) + u_l(c,l) = 0.$$

FOC for capital:

$$-u_c(c,l) + \beta \int v_k(k', z')\, dF(z'|z) = 0.$$

Envelope:

$$v_k(k,z) = u_c(c,l)(z f_k(k,l) + (1-\delta)).$$

Combine with FOC for k' to get the Euler equation:

$$u_c(c,l) = \beta \int u_c(c', l')(z' f_k(k', l') + (1-\delta))\, dF(z'|z).$$

Together with the resource constraint, these determine c, l, k'.

4 – Now assume no uncertainty and

$$u(c,l) = \log c - \frac{l^{1+\phi}}{1+\phi}, \qquad f(k,l) = zk^{\alpha}l^{1-\alpha}.$$

Compute the deterministic steady state. Explain the effect of a permanent increase in non-stochastic productivity z on capital-to-output ratio, labour, and consumption-to-output ratio, providing some economic intuition for your findings.

Solution: Given

$$u(c,l) = \log c - \frac{l^{1+\phi}}{1+\phi}, \qquad f(k,l) = k^{\alpha}l^{1-\alpha},$$

we have

$$u_c = \frac{1}{c}, \qquad u_l = -l^{\phi},$$

$$f_l = (1-\alpha)(k/l)^{\alpha}, \qquad f_k = \alpha(k/l)^{\alpha-1}.$$

Steady-state conditions:
(1) Labor condition

$$\bar{c}\,\bar{l}^{\phi} = z(1-\alpha)(\bar{k}/\bar{l})^{\alpha}.$$

(2) Euler equation

$$1 = \beta\left[z\alpha(\bar{k}/\bar{l})^{\alpha-1} + (1-\delta)\right].$$

(3) Resource constraint

$$\bar{c} + \delta\bar{k} = z\,\bar{k}^{\alpha}\bar{l}^{1-\alpha}.$$

Solve:

Capital-labor ratio:

$$\frac{\bar{k}}{\bar{l}} = \left(\frac{\alpha z}{\rho + \delta}\right)^{\frac{1}{1-\alpha}}, \qquad \rho = \frac{1}{\beta} - 1.$$

Output-labor ratio:

$$\frac{\bar{y}}{\bar{l}} = z^{1/(1-\alpha)} \left(\frac{\alpha}{\rho + \delta}\right)^{\alpha/(1-\alpha)}.$$

Capital-output ratio:

$$\frac{\bar{k}}{\bar{y}} = \frac{\alpha}{\rho + \delta}.$$

Consumption-output ratio:

$$\frac{\bar{c}}{\bar{y}} = 1 - \delta \frac{\bar{k}}{\bar{y}} = \frac{\rho + (1-\alpha)\delta}{\rho + \delta}.$$

Labor:

$$\bar{l} = \left[\frac{(1-\alpha)(\rho+\delta)}{\rho + (1-\alpha)\delta}\right]^{1/(1+\phi)}.$$

Effect of increasing z: $\bar{l}$ does not depend on z. $\bar{y}$, $\bar{k}$, $\bar{c}$ all increase more than proportionally (capital deepening). A 1% increase in z raises (c, k, y) by $1/(1-\alpha)$.

5 – Provide a sketch of pseudo code for value function iteration to solve this problem. How is uncertainty reflected in the problem of this kind? Is the presence of endogenous labour choice an issue? How can you deal with that?

Solution:

The pseudo code is standard, once one recognizes that we can substitute optimal labour choice into the objective:

$$u(c; k, z) = \log c - \frac{1}{1+\phi} \left(\left(\frac{(1-\alpha) z k^{\alpha}}{c} \right)^{\frac{1}{\phi+\alpha}} \right)^{1+\phi}$$

6 – Now write the same problem (with no uncertainty) in continuous time: specifically, write the HJB equation and derive the optimality conditions (i.e. the optimal labour condition and the Euler Equation).

Solution:

The planner solves:

$$\max_{c(t),\, l(t)} \int_0^{\infty} e^{-\rho t} \left[\log c(t) - \frac{l(t)^{1+\phi}}{1+\phi} \right] dt,$$

subject to

$$\dot{k}(t) = f(k(t), l(t)) - \delta k(t) - c(t), \qquad f(k, l) = z k^{\alpha} l^{1-\alpha}.$$

Let $V(k)$ be the value function. The Hamilton Jacobi Bellman equation is:

$$\rho V(k) = \max_{c,l} \left\{ \log c - \frac{l^{1+\phi}}{1+\phi} + V'(k)(z k^{\alpha} l^{1-\alpha} - \delta k - c) \right\}.$$

FOCs

Consumption.

$$\frac{\partial}{\partial c}: \qquad \frac{1}{c} - V'(k) = 0 \quad \Rightarrow \quad V'(k) = \frac{1}{c}. \tag{1}$$

Labor.

$$\frac{\partial}{\partial l}: \qquad -l^{\phi} + V'(k)\, z(1-\alpha)k^{\alpha} l^{-\alpha} = 0.$$

Using $V'(k) = 1/c$,

$$l^{\phi} = \frac{z(1-\alpha)k^{\alpha}}{c}\, l^{-\alpha} \quad \Rightarrow \quad c\, l^{\phi+\alpha} = z(1-\alpha)k^{\alpha}. \tag{2}$$

Envelope condition

Differentiate the HJB w.r.t. k:

$$\rho V'(k) = V'(k)\big(z f_k(k,l) - \delta\big) + V''(k)\big(z f(k,l) - \delta k - c\big).$$

Re-arranging:

$$\big(\rho - z f_k(k,l) + \delta\big) V'(k) = -\, V''(k)\big(z k^{\alpha} l^{1-\alpha} - \delta k - c\big). \tag{3}$$

Since $V'(k) = 1/c$,

$$-\frac{\dot{c}}{c} = \frac{V''(k)\, \dot{k}}{V'(k)} = \rho - z\alpha k^{\alpha-1} l^{1-\alpha} + \delta.$$

Thus the Euler equation is:

$$\frac{\dot{c}}{c} = z\alpha k^{\alpha-1} l^{1-\alpha} - \delta - \rho. \tag{4}$$

Final expressions

Labor supply rule from (2):

$$\boxed{l = \left[\frac{z(1-\alpha)}{c}\, k^{\alpha}\right]^{\frac{1}{\phi+\alpha}}.}$$

Capital accumulation:

$$\boxed{\dot{k} = z k^{\alpha} l^{1-\alpha} - \delta k - c.}$$

Consumption Euler equation:

$$\boxed{\frac{\dot{c}}{c} = z\alpha k^{\alpha-1} l^{1-\alpha} - \delta - \rho.}$$

7 – It is tedious but straightforward to show that the $\dot{c} = 0$ schedule takes the form

$$c = k^{-\phi}\left[\left(\frac{\alpha}{\delta+\rho}\right)^{\frac{1-\alpha}{\phi+\alpha}} (1-\alpha) z^{\frac{1+\phi}{\phi+\alpha}}\right].$$

Taking this result as given (so no need to derive it), draw the phase diagram, assuming the $\dot{k} = 0$ equation is given by the usual $c = zk^{\alpha} - \delta k$ (i.e. assume that ϕ is large and so labour supply is almost inelastic, so its effect on the $\dot{k} = 0$ can be ignored (otherwise it's a bit messy!)).

Trace out the dynamics following a permanent shock to productivity z.

Solution:

$$\frac{\dot{c}}{c} = z\alpha k^{\alpha-1}\left[\frac{z(1-\alpha)}{c}\, k^{\alpha}\right]^{\frac{1-\alpha}{\phi+\alpha}} - \delta - \rho.$$

$$\frac{\dot{c}}{c} = z\alpha k^{\left((\alpha-1)+\alpha\frac{1-\alpha}{\phi+\alpha}\right)}\left[\frac{z(1-\alpha)}{c}\right]^{\frac{1-\alpha}{\phi+\alpha}} - \delta - \rho.$$

$$\frac{\dot{c}}{c} = z\alpha k^{(1-\alpha)\left(\frac{\alpha}{\phi+\alpha}-1\right)} \left[\frac{z(1-\alpha)}{c}\right]^{\frac{1-\alpha}{\phi+\alpha}} - \delta - \rho.$$

$$\frac{\dot{c}}{c} = z\alpha k^{-(1-\alpha)\left(\frac{\phi}{\phi+\alpha}\right)} \left[\frac{z(1-\alpha)}{c}\right]^{\frac{1-\alpha}{\phi+\alpha}} - \delta - \rho.$$

$$\left(\frac{\delta+\rho}{z\alpha}\right)(z(1-\alpha))^{-\frac{1-\alpha}{\phi+\alpha}} = k^{-(1-\alpha)\left(\frac{\phi}{\phi+\alpha}\right)} \left[\frac{1}{c}\right]^{\frac{1-\alpha}{\phi+\alpha}}.$$

$$\left(\frac{\delta+\rho}{z\alpha}\right)^{\frac{\phi+\alpha}{1-\alpha}}(z(1-\alpha))^{-1} = k^{-\phi}\left[\frac{1}{c}\right].$$

$$c = k^{-\phi}\left[\left(\frac{\alpha z}{\delta+\rho}\right)^{\frac{1-\alpha}{\phi+\alpha}}(1-\alpha)z\right].$$

$$c = k^{-\phi}\left[\left(\frac{\alpha}{\delta+\rho}\right)^{\frac{1-\alpha}{\phi+\alpha}}(1-\alpha)z^{\frac{1+\phi}{\phi+\alpha}}\right].$$

Thus, in the (k, c) phase diagram, the $\dot{c} = 0$ locus is an downward sloping curve. Dynamics are as in the lectures, but students need to work this out by inspecting the Euler Equation. Best answers would recognize that from the phase diagram alone it is impossible to tell whether consumption increases or decreases on impact - can be drawn both ways. Below is the case when consumption is unchanged on impact and a case where it drops.

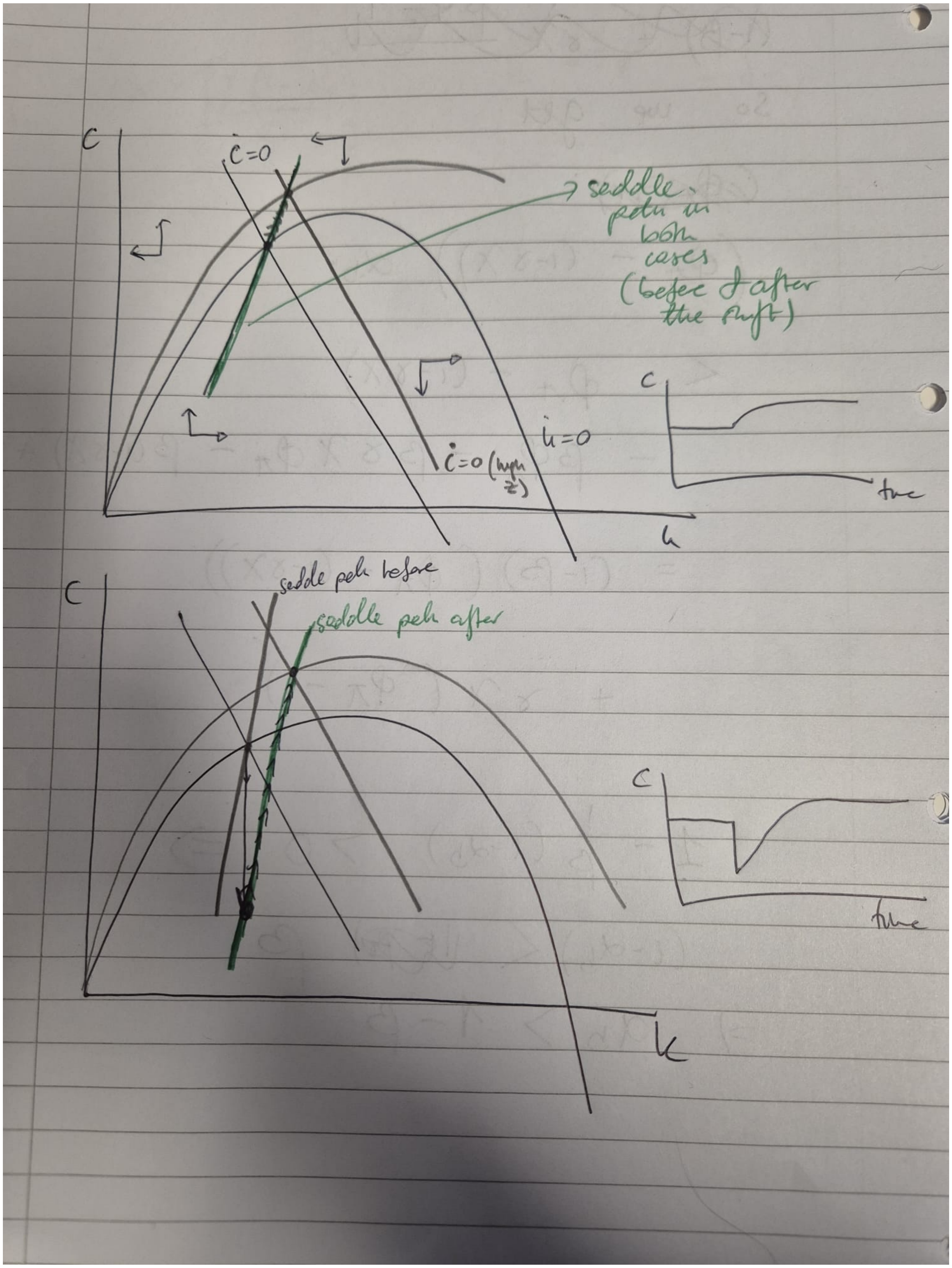

Figure 1: Phase diagram

IV – Two countries in the Aiyagari model

Consider the stationary equilibrium of the Aiyagari model covered in class. The representative firm's production function is $Y = ZK^{\alpha}N^{1-\alpha}$ with $\alpha \in (0,1)$ and where K and N are the aggregate quantities of capital and labor and Z is aggregate productivity. Consider two countries, A and B, which are identical in all respects except for their inhabitant's discount rates, in particular $\rho_A > \rho_B$. Denoting the stationary equilibrium interest rates by r_A^* and r_B^* and the stationary capital stocks by K_A^* and K_B^* how do you expect these to differ across the two countries when the countries are in autarky? What will happen to their interest rates, capital stocks, and net foreign asset positions when these countries integrate financially? Draw a diagram to illustrate.

Solution: In autarky, $r_A > r_B$. When countries integrate, the country with lower discount rate builds a positive NFA by lending to the other country. Its domestic capital stock decreases. The opposite is true in country A.

End of Exam